

Homework 5

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30. (a)

$$\begin{aligned} f_X(x) &= \begin{cases} \int_0^{1-x} 60x^2y \, dy & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases} \\ &= \begin{cases} 30x^2(1-x)^2 & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}. \end{aligned}$$

$$\begin{aligned} f_Y(y) &= \begin{cases} \int_0^{1-y} 60x^2y \, dx & 0 < y < 1 \\ 0 & \text{otherwise} \end{cases} \\ &= \begin{cases} 20(1-y)^3y & 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}. \end{aligned}$$

(b) From the exponents in the marginal distributions, we see $X \sim \text{Beta}(3, 3)$, $Y \sim \text{Beta}(2, 4)$. The mean and variance formulas for $Z \sim \text{Beta}(\alpha, \beta)$ are

$$\begin{aligned} E(Z) &= \frac{\alpha}{\alpha + \beta}, \\ V(Z) &= \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}. \end{aligned}$$

Plug in to obtain

$$\begin{aligned} E(X) &= 1/2, \\ E(Y) &= 1/3, \\ V(X) &= \frac{9}{36 \cdot 7} = \frac{1}{28}, \\ V(Y) &= \frac{8}{36 \cdot 7} = \frac{2}{63}. \end{aligned}$$

(c)

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y),$$

where

$$\begin{aligned}
E(XY) &= \iint_{x>0, y>0, x+y<1} xy(60x^2y) dx dy \\
&= \int_0^1 \int_0^{1-x} 60x^3y^2 dy dx \\
&= 20 \int_0^1 x^3(1-x)^3 dx \\
&= -20 \int_0^1 x^6 - 3x^5 + 3x^4 - x^3 dx \\
&= -20 \left(\frac{1}{7} - \frac{3}{6} + \frac{3}{5} - \frac{1}{4} \right) \\
&= \frac{1}{7}.
\end{aligned}$$

Since $E(X)E(Y) = 1/6$, we obtain $\text{Cov}(X, Y) = 1/7 - 1/6 = 1/42$. Thus

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{V(X)V(Y)}} = \frac{-\frac{1}{42}}{\sqrt{\frac{1}{28} \frac{2}{63}}} = -\frac{\sqrt{2}}{2}.$$

(d)

$$V(X + Y) = V(X) + V(Y) + 2\text{Cov}(X, Y) = \frac{1}{28} + \frac{2}{63} - \frac{1}{21} = \frac{5}{252}.$$

32. Let

$$\begin{aligned}
f_{X_1}(x_1) &= \begin{cases} e^{-x_1} & x_1 > 0 \\ 0 & \text{otherwise} \end{cases}, \\
f_{X_2}(x_2) &= \begin{cases} 2e^{-2x_2} & x_2 > 0 \\ 0 & \text{otherwise} \end{cases}, \\
f_{X_3}(x_3) &= \begin{cases} 3e^{-3x_3} & x_3 > 0 \\ 0 & \text{otherwise} \end{cases}.
\end{aligned}$$

We will prove that they are marginal pdf of X_1, X_2 and X_3 , respectively. First, it is easy to see that $f_{X_1, X_2, X_3}(x_1, x_2, x_3) = f_{X_1}(x_1)f_{X_2}(x_2)f_{X_3}(x_3)$ and

$$\int_{-\infty}^{\infty} f_{X_1}(x_1) dx_1 = \int_{-\infty}^{\infty} f_{X_2}(x_2) dx_2 = \int_{-\infty}^{\infty} f_{X_3}(x_3) dx_3 = 1.$$

Thus the marginal pdf of X_1 by definition is

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X_1, X_2, X_3}(x_1, x_2, x_3) dx_2 dx_3 = f_{X_1}(x_1) \int_{-\infty}^{\infty} f_{X_2}(x_2) dx_2 \int_{-\infty}^{\infty} f_{X_3}(x_3) dx_3 = f_{X_1}(x_1).$$

Similarly, we can conclude that the marginal pdf of X_2 is $f_{X_2}(x_2)$, and the marginal pdf of X_3 is $f_{X_3}(x_3)$. Since the joint density function is equal to the product of the marginal pdfs ($f_{X_1, X_2, X_3}(x_1, x_2, x_3) = f_{X_1}(x_1)f_{X_2}(x_2)f_{X_3}(x_3)$), we know X_1, X_2 and X_3 are independent.

34. (a)

$$\begin{aligned}
f_{X_1, X_2}(x_1, x_2) &= \int_{-\infty}^{\infty} f_{X_1, X_2, X_3}(x_1, x_2, x_3) dx_3 \\
&= \begin{cases} \int_0^1 \frac{1}{3}(x_1 + 2x_2 + 3x_3) dx_3 & 0 < x_i < 1, i = 1, 2 \\ 0 & \text{otherwise} \end{cases} \\
&= \begin{cases} \frac{1}{3}(x_1 + 2x_2) + \frac{1}{2} & 0 < x_i < 1, i = 1, 2 \\ 0 & \text{otherwise} \end{cases}.
\end{aligned}$$

(b) Since

$$\begin{aligned}
f_{X_3|X_1=\frac{1}{4}, X_2=\frac{3}{4}}(x_3) &= \frac{f_{X_1, X_2, X_3}\left(\frac{1}{4}, \frac{3}{4}, x_3\right)}{f_{X_1, X_2}\left(\frac{1}{4}, \frac{3}{4}\right)} \\
&= \begin{cases} \frac{\frac{1}{3}\left(\left(\frac{1}{4}\right)+2\left(\frac{3}{4}\right)+3x_3\right)}{\frac{1}{3}\left(\left(\frac{1}{4}\right)+2\left(\frac{3}{4}\right)\right)+\frac{1}{2}} & 0 < x_3 < 1 \\ 0 & \text{otherwise} \end{cases} \\
&= \begin{cases} \frac{12x_3+7}{13} & 0 < x_3 < 1 \\ 0 & \text{otherwise} \end{cases},
\end{aligned}$$

we have

$$\begin{aligned}
P\left(X_3 < \frac{1}{2} \middle| X_1 = \frac{1}{4}, X_2 = \frac{3}{4}\right) &= \int_0^{\frac{1}{2}} f_{X_3|X_1=\frac{1}{4}, X_2=\frac{3}{4}}(x_3) dx_3 \\
&= \int_0^{\frac{1}{2}} \frac{12x_3+7}{13} dx_3 \\
&= \frac{5}{13}.
\end{aligned}$$

42. (a) Taking away all terms related to X_1 , we see that (X_2, X_3) follows a bivariate normal distribution with mean $(2, 0)^T$ and covariance matrix

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

from which we see that $\text{Cov}(X_2, X_3) = 0$ and hence X_2 and X_3 are independent.

- (c) Since X is multivariate normal, so is $Y = AX$. Its mean is

$$A\mu = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix},$$

and its covariance matrix is

$$A\Sigma A^T = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 & \frac{1}{2} \\ 0 & 1 & 0 \\ \frac{1}{2} & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 0 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 3 & \frac{9}{2} \\ \frac{9}{2} & 7 \end{pmatrix}.$$

Since the covariance between the two components is $\frac{9}{2}$, not zero, the components are dependent.